

Elif Nur Tekay

Software Engineering | Python, C++, Java, LLM Evaluation & Test Automation

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About

Software Engineering graduate and thesis-based M.Sc. student in Software Engineering at Kocaeli University. I have academic and hands-on project experience in software development with C++, Python, and Java, as well as object-oriented programming, software/model integration, artificial intelligence, machine learning, computer vision, and image processing. My capstone project, supported through the TUSAŞ LIFT UP Program, won first place in the Artificial Intelligence and Data Science category at Ankara University's IT Project Market and Career Days, and the resulting research received the Best Student Paper Award at INFUS 2026. Our TÜBİTAK 2209-A-supported MemoVision project placed second among 829 teams in Turkcell's Tomorrow's Technology Leaders Competition. Through multidisciplinary projects, I have gained experience in research, software development, system integration, testing, performance evaluation, and technical documentation. I am particularly interested in software engineering, LLM and coding-agent evaluation, automated testing, and systematic evaluation of AI-enabled software systems.

Education

Kocaeli University

M.Sc. in Software Engineering, Thesis Program

Malatya Turgut Özal University

B.Sc. in Software Engineering; GPA: 3.37/4.00

Ankara University

Software Engineering, Special Student Status

Kocaeli, Türkiye

2026 – Present

Malatya, Türkiye

2022 – 2026

Ankara, Türkiye

2023 – 2025

Experience

ANLAM-NET Research LAB, Ankara University

Research Assistant

Ankara, Türkiye

Aug. 2024 – Present

- Contribute to AI and multimodal research projects involving data preprocessing, feature extraction, model development, experimental evaluation, performance analysis, and software/model integration.
- Work in collaborative Git/GitHub-based workflows involving experiment planning, debugging, technical documentation, sprint planning, and task tracking using Agile/Scrum practices.

TÜBİTAK SAGE

Artificial Intelligence Applications Intern

Ankara, Türkiye

Jul. 2025 – Aug. 2025

- Fine-tuned and evaluated deep-learning super-resolution models for large-scale satellite imagery using patch-based processing, stitching, and comparative performance analysis.
- Contributed to model validation, optimization, output analysis, and technical documentation in an applied AI research environment.

Ankara University Project Support Office

Part-Time Student Employee

Ankara, Türkiye

Mar. 2025 – Jun. 2025

- Managed website content and updates, and supported documentation and progress tracking for academic project processes.

SA Innovation, Hacettepe Teknokent

Intern

Ankara, Türkiye

Jul. 2024 – Aug. 2024

- Prepared audio and text datasets for an AI-based sentiment analysis project through data cleaning, normalization, labeling, and quality control.

Technical Skills

Programming: Python, Java, C++, C, SQL, JavaScript

AI/ML: PyTorch, Scikit-learn, Hugging Face Transformers, Deep Learning, Machine Learning, Computer Vision, Vision Transformers, Multimodal AI, Generative AI, Reinforcement Learning, XAI

LLM & AI Evaluation: LLM Integration, Agent-Based Automation, Model Evaluation, RAG, Document Retrieval

Data & Experimentation: Pandas, NumPy, Data Preprocessing, Feature Extraction, Performance Analysis, Experimental Validation

Software Engineering: Model Integration, Automated Testing, Test Scenario Design, Debugging, Inference Workflows, Proof-of-Concept Development, REST API, FastAPI, Flask, Django, CI Workflows

Databases: PostgreSQL, MySQL, MongoDB, SQLite3

Tools: Git, GitHub, GitHub Actions, Docker, Linux, OpenCV, Jira, Asana, Postman, VS Code, Google Colab

Languages: Turkish (Native), English (B2 – Upper-Intermediate; academic writing and international presentation experience)

Projects

TraceForge – LLM-Based Software Engineering Evaluation Framework

2026

Coding-Agent Evaluation, Benchmarking & Reliability Analysis

- Developed a reproducible framework for evaluating software-engineering agent trajectories, detecting policy and workflow failures, and identifying the first erroneous step with constraint-level supporting evidence.
- Built a 15-case benchmark with development and held-out test splits, baseline/advanced evaluation modes, structured JSON outputs, regression checks, evidence-quality metrics, retry handling, latency/token tracking, and automated comparison reports.

- Developed and evaluated reinforcement-learning-based aerodynamic optimisation workflows using PPO, TD3, and SAC with CST parameterisation, hybrid surrogate models, and XFOIL/NeuralFoil validation.
- Contributed to reward design, engineering constraints, model validation, performance comparison, SHAP-based explainability, and software integration. The project won first place in Ankara University’s Artificial Intelligence and Data Science category and resulted in the INFUS 2026 Best Student Paper Award.

- Developed multimodal machine-learning approaches combining visual, temporal, textual, and metadata features using CLIP, MLP, Vision Transformer, classical ML, and deep-learning methods across multiple video datasets.
- Performed feature extraction, multimodal fusion, model comparison, experimental validation, and performance analysis, and contributed to integrating trained models into a working web application. MemoVision placed second among 829 teams in Turkcell’s competition.

- Contribute to Python-based AI workflows for UAV and helicopter object detection and tracking, camera-based position estimation, reference-object detection, inference, model integration, evaluation, and output validation.
- Work on performance analysis and integration of computer-vision and software components in a collaborative Git/GitHub-based development environment.

- Built machine learning pipelines for mental-state inference from multimodal physiological signals using preprocessing, feature extraction, subject-wise data splitting, and multiple regression/classification models.
- Compared classical ML and deep-learning approaches, including XGBoost, KNN, SVM, MLP, CNN-1D, GRU, and Transformer Fusion, with hyperparameter optimization and performance evaluation.

- Contributed to a real-time and offline heart-sound analysis application supporting waveform visualisation, peak detection, BPM calculation, and anomaly-status reporting; improved modularity and testability through OOP and Clean Code refactoring.

- Developed a full-stack project management application supporting task assignment, team management, progress and deadline tracking, and Gantt chart visualization; integrated a Django/MongoDB backend with a React-based user interface.

- Developed a web-based communication application supporting real-time messaging, voice calls, and video calls using Node.js, WebSocket, and WebRTC; worked on server-side communication and MySQL database integration.

Publications

Comparative Analysis of MLP and Vision Transformers for Video Memorability Classification <i>IEEE UBMK 2025 – IEEE Xplore, DOI: 10.1109/UBMK67458.2025.11207010</i>	2025
Multimodal Feature Fusion for Video and Brand Memorability <i>MediaEval 2025 Workshop Proceedings – Paper Link</i>	2025
Reinforcement Learning-Based Airfoil Shape Optimization Using Hybrid Surrogate Models <i>INFUS 2026 – Presented; Proceedings Publication in Progress</i>	2026
A CLIP-Based Visual-Semantic Embedding Approach for Commercial Memorability Prediction <i>MediaEval 2026 Workshop Proceedings – Paper Link</i>	2026

Awards

Best Student Paper Award <i>8th International Conference on Intelligent and Fuzzy Systems (INFUS 2026)</i>	Jul. 2026
First Place – Artificial Intelligence and Data Science Category <i>Ankara University IT Project Market and Career Days</i>	May 2026
Second Place – MemoVision <i>Turkcell Tomorrow’s Technology Leaders Competition – 829 teams</i>	May 2026

Certificates

Data Analysis School 2025–2026 – Artificial Intelligence and Machine Learning (Council of Higher Education – YÖK) | Deep Learning Bootcamp | Machine Learning Bootcamps (Akbank & Aygaz) | ROKETSAN Level Up AI Workshop – Developing an AI Agent: Document-Aware Intelligent Assistant | Python Programming 401 | C++ Programming (BTK Academy) | Java Programming (BTK Academy) | Web Development with Django (BTK Academy) | White Hat Hacker (Siber Vatan) | Introduction to Linux and Networks (Cisco)

References